Wireshark and OSI Model

1. Capture Network Traffic Using Wireshark

Steps

1. Download and install Wireshark from the official website: https://www.wireshark.org/download.html
2. Open Wireshark and select your active Wi-Fi or Ethernet interface.
3. Double-click the interface to start capturing packets.
4. Open a browser and visit https://www.instagram.com/.
5. Wait a few seconds, then return to Wireshark and click the red Stop button.
6. Look at the Protocol column in the packet list and record the protocols you see.

Sample observation table

These are common examples; your actual capture may show different protocols.

No.	Protocol	Purpose
1	DNS	Resolves domain names to IP addresses.
2	TCP	Provides reliable transport for many connections.
3	TLSv1.3	Helps encrypt HTTPS traffic.
4	QUIC	Provides transport functionality used by HTTP/3.
5	HTTP	Transfers web requests and responses; HTTPS traffic may not display as plain HTTP.
6	ARP	Resolves IPv4 addresses to MAC addresses on a local network.
7	ICMPv6	Supports IPv6 network control and diagnostics.

Record only the protocols that actually appear in your capture.

2. Identify Three Packets and Their OSI Layers

Use three packets from your actual Wireshark capture. Here is a sample table you can adapt to match your results.

Packet	Protocol	OSI Layer	How I Identified It
1	DNS	Application (Layer 7)	The packet details show a DNS query for a domain name.
2	TCP	Transport (Layer 4)	The header contains source and destination port numbers, along with TCP flags.

3	TLS 1.3	Mainly Presentation (Layer 6) conceptually	TLS provides encryption and data protection for the application connection.

Note: OSI is a conceptual model, so TLS does not map perfectly to a single layer. In a real capture, you may see DNS, TCP, TLS or QUIC depending on the website and network.

3. Colorful OSI Model Diagram

The generated infographic above includes the seven OSI layers, real-world examples, and a WhatsApp message flow. Use its OSI Model panel as a visual reference for your assignment.

The seven layers, from top to bottom, are:

1. Application — WhatsApp, Instagram, web browsing
2. Presentation — encryption and data formatting
3. Session — managing communication sessions
4. Transport — TCP and UDP
5. Network — IP addressing and routing
6. Data Link — Ethernet frames and MAC addresses
7. Physical — cables, radio waves and electrical signals

4. How a WhatsApp Message Travels Through the OSI Model

1. Application Layer: I type a message in WhatsApp and press Send.
2. Transport Layer: Transport protocols carry the message data between endpoints; TCP can split data into segments and provide reliable delivery.
3. Network Layer: IP addresses are used to route packets toward the destination.
4. Data Link Layer: The packet is placed inside a frame with local-link addressing so it can travel through my Wi-Fi network.
5. Physical Layer: The frame is transmitted as radio signals over Wi-Fi or electrical signals over Ethernet.

At the receiving end, the data passes back up the layers until the application can process and display the message.